



Online PH Sensor Modbus Protocol

MODBUS RTU Transmission Mode

Encoding system	8-bit binary Each 8-bit byte in the protocol contains two 4-bit hexadecimal characters (0-9, A-F)			
Bit per byte	1 start bit 8 data bits, the least significant bit is sent first No parity check bit 1 stop bit			
Baud rate	9600bps			
Set slave ID 01 to 05	SEND: 01 10 08 00 00 01 02 05 00 2C C0 GET: 01 10 08 00 00 01 03 A9			
Get pH pH is 6.95	01 03 06 00 00 02 C4 83 01 03 04 63 7A DE 40 9D FE			
Start pH calibration	The pH electrode can support three-point calibration. The standard buffer solution is			
	4.00	6.86	9.18	(25℃)
The calibration sequence of the three-point calibration is that the first point is calibrated at pH 4, the second point is calibrated at pH 6.86, and the third point is calibrated in a buffer solution of pH 9.18. The calibration process is: Step 1: Immerse the electrode in the first calibration standard for at least 1 minute. Step 2: Start calibration: the user enters the pH value of the calibration standard 01 10 12 00 00 02 04 00 00 80 40 CRC 01 10 12 00 00 02 4A 4C Step 3: Repeat the above two steps to calibrate the second and third points 01 10 12 00 00 02 04 1F 85 DB 40 6A 32 01 10 12 00 00 02 44 B0 01 10 12 00 00 02 04 48 E1 12 41 AC 09 01 10 12 00 00 02 44 B0 Step 4: Read the calibration status and judge whether the calibration is successful.				
Criteria for successful calibration	The pH value is stable for more than 1 minute; The currently measured pH value matches one of the calibration standard solutions; The calibration parameters should be within a reasonable range.			
Get temperature value 18.75	01 03 07 00 00 02 C5 7F 01 03 04 2D 08 96 41 DD 0D。			
Set pH user calibration parameters K=1 B=0	SEND: 01 10 09 00 00 04 08 00 00 80 3F 00 00 00 00 A0 76 GET: 01 10 09 00 00 04 C2 56			
Get pH user calibration parameters K=1 B=0	SEND: 01 03 09 00 00 04 47 95 GET: 01 03 08 00 00 80 3F 00 00 00 00 9E 12			



Get slave ID **01**

SEND: FF 03 08 00 00 01 93 B4

GET: FF 03 02 **01** 00 90 00